ORDINARY DIFFERENTIAL EQUATIONS • UNIT 3.01

C Second-Order Linear Homogeneous ODEs & Wronskian

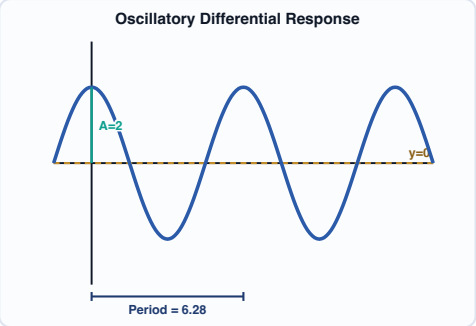
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Second-Order Linear Homogeneous ODEs & Wronskian Practice Exercises

1 MCQ

The second-order linear homogeneous ODE $f''(x) - 39f'(x) + 350f(x) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.



A 39

B 24

C 42

D 53

2 MCQ

Consider the linear first-order differential equation $\frac{dy}{dx} + 2y = 6$. If the initial condition is $y(0) = 1$, find the value of $y(1)$.

A 2

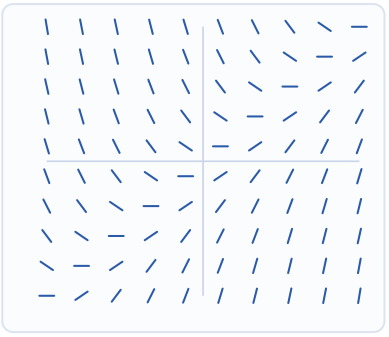
B $e^2 - 1$

C $3 - 2e^{-2}$

D $3 - 2e^{-1}$

3 MCQ

Consider the differential equation $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. Use the Euler method with a step size of $h = 0.1$ to approximate $y(0.2)$.



A 1.200

B 1.210

C 1.230

D 1.220

4 MCQ

Consider the second-order linear homogeneous differential equation $y'' - 5y' + 6y = 0$. If the auxiliary equation has roots r_1 and r_2, find the value of $r_1 + r_2$.

A 5

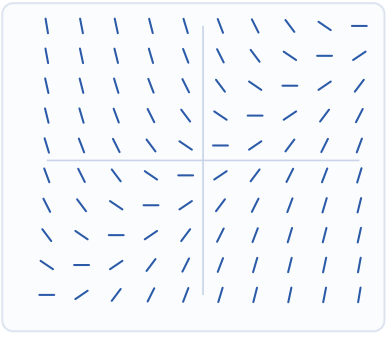
B -5

C 6

D -6

5 MCQ

Consider the separable differential equation $\frac{dy}{dx} = 2x$ with the initial condition $y(0) = 3$. Find the value of $y(2)$.



A 7

B 6

C 8

D 5

6 MCQ

Consider the differential equation $y'' - xy = 0$. The point $x = 0$ is an ordinary point. If we assume a power series solution of the form $y(x) = \sum_{n=0}^{\infty} c_n x^n$, find the recurrence relation for the coefficients c_n.

A $c_{n+2} = \frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$

B $c_{n+2} = \frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$

C $c_{n+2} = -\frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$

D $c_{n+2} = -\frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$

7 MCQ

The second-order linear homogeneous ODE $u''(s) - 40u'(s) + 144u(s) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.

A 44

B 43

C 24

D 40

8 MCQ

The Laplace transform of $u(s) = 170e^{19s}$ is $v(s) = \dfrac{170}{s - 19}$ for $s > 19$. Determine the evaluated transform $v(20)$.

A 170

B 151

C 173

D 187

9 MCQ

The second-order linear homogeneous ODE $F''(w) - 50F'(w) + 621F(w) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.

A 31

B 73

C 50

D 53

10 MCQ

Consider the second-order linear homogeneous differential equation $y'' - 2y' + y = 0$.
If one of the fundamental solutions is $y_1(x) = e^x$, and the Wronskian of two linearly independent solutions $y_1(x)$ and $y_2(x)$ is given by $W(y_1, y_2)(x) = 5e^{2x}$, find the unknown parameter k in $y_2(x) = kxe^x$.

A $k = -5$

B $k = 1$

C $k = 5$

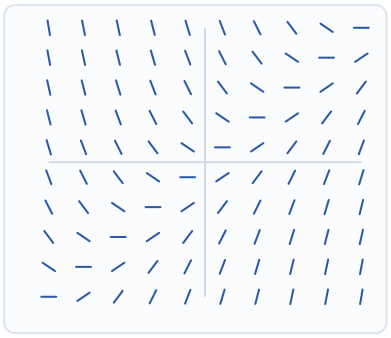
D $k = -1$

11 MCQ

Consider the second-order linear homogeneous differential equation

$$y'' - 3y' + 2y = 0.$$

Find the general solution to this equation.



A $y(x) = c_1 e^{-x} + c_2 e^{-2x}$

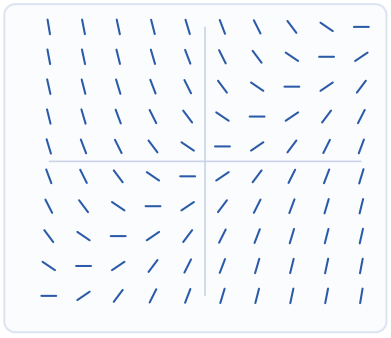
B $y(x) = c_1 e^{x} + c_2 e^{2x}$

C $y(x) = c_1 \cos(x) + c_2 \sin(x)$

D $y(x) = c_1 x e^{x} + c_2 e^{x}$

12 MCQ

Consider the second-order linear nonhomogeneous differential equation $y'' - y = \sin x$. If $y_p(x) = A\cos x + B\sin x$ is a particular solution, find the values of the constants A and B.



A $A = \frac{1}{2}, B = 0$

B $A = 0, B = \frac{1}{2}$

C $A = -\frac{1}{2}, B = 0$

D $A = 0, B = -\frac{1}{2}$

13 MCQ

Consider the second-order linear homogeneous differential equation $y'' - 2y' + ky = 0$. We are given that $y_1(x) = e^{3x}$ is a solution to this equation.

(a) Find the value of the parameter k.

(b) Find the general solution to the differential equation.

A $k = 3, y(x) = c_1 e^{3x} + c_2 x e^{3x}$

B $k = -3, y(x) = c_1 e^{3x} + c_2 e^{x}$

C $k = 9, y(x) = c_1 e^{3x} + c_2 e^{-3x}$

D $k = -3, y(x) = c_1 e^{3x} + c_2 e^{-x}$

14 MCQ

Consider the second-order linear non-homogeneous differential equation: $\frac{d^2y}{dx^2} + 4y = \sec(2x)$. Suppose a particular solution is found using the method of variation of parameters. Let the fundamental set of solutions for the associated homogeneous equation be $y_1(x) = \cos(2x)$ and $y_2(x) = \sin(2x)$. If the particular solution is expressed in the form $y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$, determine the value of $u_2'(x)$.

A $\frac{1}{4}$

B $\frac{1}{2}$

C $\frac{1}{2}\sin(2x)$

D $\frac{1}{4}\cos(2x)$

15 MCQ

Consider the second-order linear homogeneous differential equation $y''(x) + 5y'(x) + 6y(x) = 0$. If the auxiliary equation has roots r_1 and r_2, what is the value of $r_1 + r_2$?

A -5

B -6

C 5

D 6

16 MCQ

The separable initial value problem $\frac{dp}{dt} = 34t$ satisfies $p(0) = 5$. Determine the solution value $p(18)$.

A 5530

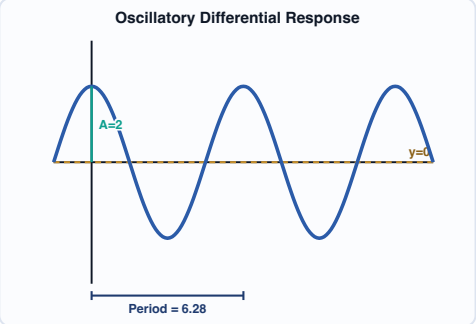
B 5508

C 5516

D 5513

17 MCQ

The second-order linear homogeneous ODE $p''(t) - 41p'(t) + 210p(t) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.



A 41

B 21

C 47

D 44

18 MCQ

The Laplace transform of $p(t) = 6e^{17t}$ is $q(s) = \dfrac{6}{s-17}$ for $s > 17$. Determine the evaluated transform $q(18)$.

A -11

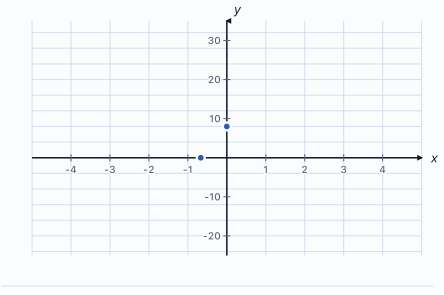
B 6

C 9

D 8

19 MCQ

Consider the nonhomogeneous second-order linear differential equation $y'' - 5y' + 6y = 12x + 8$. $y_h(x) = c_1e^{2x} + c_2e^{3x}$ is the general solution to the corresponding homogeneous equation. Find the particular solution $y_p(x)$ using the method of undetermined coefficients, and then determine the value of the constant A such that $y_p(x) = Ax + B$ for some constant B.



A $A = 3$

B $A = 2$

C $A = 6$

D $A = 12$

20 MCQ

The second-order linear homogeneous ODE $h''(u) - 42h'(u) + 272h(u) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.

A 45

B 50

C 23

D 42

21 MCQ

Consider the second-order linear homogeneous differential equation: $x^2y'' + xy' + (x^2 - \nu^2)y = 0$, where ν is a constant. This is Bessel's equation of order ν. Assume $\nu = 1/2$ and that $x > 0$.

Part 1: Ordinary Point Analysis

$x = 0$ is a regular singular point. Determine the indicial equation for this differential equation and find its roots.

A Indicial equation: $r^2 + \frac{1}{4} = 0$; roots: $r = \pm\frac{i}{2}$

B Indicial equation: $r^2 - r + \frac{1}{4} = 0$; roots: $r = \frac{1}{2}, \frac{1}{2}$

C Indicial equation: $r^2 - \frac{1}{4} = 0$; roots: $r = \pm\frac{1}{4}$

D Indicial equation: $r^2 - \frac{1}{4} = 0$; roots: $r = \pm\frac{1}{2}$

22 MCQ

The second-order linear homogeneous ODE $u''(s) - 43u'(s) + 460u(s) = 0$ has characteristic roots λ_1, λ_2. Compute the trace sum $\lambda_1 + \lambda_2$.

A 32

B 46

C 66

D 43

23 MCQ

Consider the nonhomogeneous second-order linear differential equation $y'' + 4y = 8$. What is the particular solution, $y_p(x)$, to this equation?

A 2

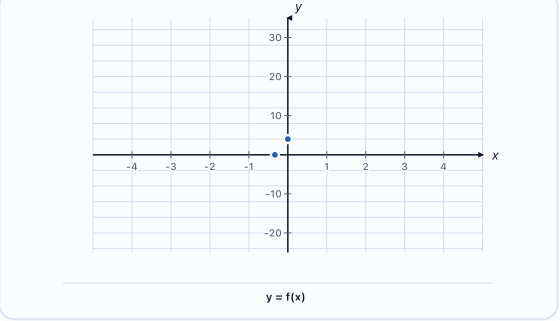
B $2x$

C $2\sin(2x)$

D $2\cos(2x)$

24 MCQ

Consider the second-order nonhomogeneous linear differential equation $y'' - 5y' + 6y = 12x + 4$. If a particular solution is of the form $y_p(x) = Ax + B$ and the general solution is $y(x) = C_1e^{2x} + C_2e^{3x} + Ax + B$, find the value of A.



A 1

B 6

C 12

D 2

25 MCQ

The second-order linear homogeneous ODE $M''(z) - 49M'(z) + 510M(z) = 0$ has characteristic roots λ_1, λ_2. Compute the trace sum $\lambda_1 + \lambda_2$.

A 64

B 52

C 33

D 49